String Methods

In [4]:
```
'hello'
```
Out[4]:
```
'hello'
```

In [3]:
```
"hello"
```
Out[3]:
```
'hello'
```

In [10]:
```
"""hello"""

a = 5
a
```
Out[10]:
```
5
```

In [13]:
```
a = 6
b = 3

"""aaa"""

a, b
```
Out[13]:
```
(6, 3)
```

In [ ]:
```

```

In [1]:
```
string = 'training basket'
string
```
Out[1]:
```
'training basket'
```

In [2]:
```
type(string)
```
Out[2]: str

In [3]:
```
s = string
```

In [4]:
```
s
```
Out[4]: 'training basket'

In [5]:
```
len(s)
```
Out[5]: 15

In [8]:
```
s.capitalize()
```
Out[8]: 'Training basket'

In [9]:
```
s.title()
```
Out[9]: 'Training Basket'

In [10]:
```
s.upper()
```
Out[10]: 'TRAINING BASKET'

In [11]:
```
s1 = 'TRAINING BASKET'
s1
```
Out[11]: 'TRAINING BASKET'

In [12]:
```
s1.lower()
```
Out[12]: 'training basket'

In [13]:
```
s
```
Out[13]: 'training basket'

In [21]:
```
s.replace(' ', '--')
```
Out[21]: 'training--basket'

In [ ]:
```

```

In [22]:
```
s
```
Out[22]: 'training basket'

In [23]:
```
s.split()
```
Out[23]: ['training', 'basket']

In [24]:
```
s2 = 'training--basket'
s2
```
Out[24]: 'training--basket'

In [29]:
```
s2.split(sep='--')
```
Out[29]: ['training', 'basket']

In [31]:
```
date = '01-22-2024'
date.split('-')
```
Out[31]: ['01', '22', '2024']

In [33]:
```
date.replace('-', '/')
```
Out[33]: '01/22/2024'

In [34]:
```
s
```
Out[34]: 'training basket'

In [39]:
```
s.count('i')
```
Out[39]: 2

In [40]:
```
s.index('a')
```
Out[40]: 2

In [41]:
```
s
```
Out[41]: 'training basket'

In [43]:
```
s.find('ing')
```
Out[43]: 5

In [44]:
```
x = '246xyz'
x
```
Out[44]: '246xyz'

In [45]:
```
x.isalnum()
```
Out[45]: True

In [46]:
```
x.isalpha()
```
Out[46]: False

In [47]:
```
x.isnumeric()
```
Out[47]: False

In [48]:
```
y = '1234'
y.isnumeric()
```
Out[48]: True

In [49]:
```
z = 'abcd'
z.isalpha()
```
Out[49]: True

In [51]:
```
y.isalnum()
```
Out[51]: True

In [ ]:
```

```

In [52]:
```
e = 'ESC'
e
```
Out[52]: 'ESC'

In [53]:
```
e.isascii()
```
Out[53]: True

In [54]:
```
'ctrl'.isascii()
```
Out[54]:
```
True
```

In [55]:
```
'k'.isascii()
```
Out[55]:
```
True
```

In [ ]:
```
```

List Data Types

In [56]:
```
lst = [2,3,6,4,2,4,6]
lst
```
Out[56]:
```
[2, 3, 6, 4, 2, 4, 6]
```

In [57]:
```
type(lst)
```
Out[57]:
```
list
```

List Methods

In [58]:
```
lst
```
Out[58]:
```
[2, 3, 6, 4, 2, 4, 6]
```

In [59]:
```
lst.append(25)
```

In [60]:
```
lst
```
Out[60]: [2, 3, 6, 4, 2, 4, 6, 25]

In [61]:
```
lst.append(3.45)
lst
```
Out[61]: [2, 3, 6, 4, 2, 4, 6, 25, 3.45]

In [62]:
```
lst.append([2,5,3,4])
lst
```
Out[62]: [2, 3, 6, 4, 2, 4, 6, 25, 3.45, [2, 5, 3, 4]]

In [63]:
```
lst.extend([2,5,3,4])
lst
```
Out[63]: [2, 3, 6, 4, 2, 4, 6, 25, 3.45, [2, 5, 3, 4], 2, 5, 3, 4]

In [ ]:
```

```

In [64]:
```
lst.pop()
```
Out[64]: 4

In [65]:
```
lst
```
Out[65]: [2, 3, 6, 4, 2, 4, 6, 25, 3.45, [2, 5, 3, 4], 2, 5, 3]

In [66]:
```
lst.pop(9)
lst
```
Out[66]: [2, 3, 6, 4, 2, 4, 6, 25, 3.45, 2, 5, 3]

In [67]:
```
lst.remove(3.45)
lst
```
Out[67]: [2, 3, 6, 4, 2, 4, 6, 25, 2, 5, 3]

In [68]:
```
lst
```
Out[68]: [2, 3, 6, 4, 2, 4, 6, 25, 2, 5, 3]

In [69]:
```
lst.reverse()
lst
```
Out[69]: [3, 5, 2, 25, 6, 4, 2, 4, 6, 3, 2]

In [70]:
```
lst
```
Out[70]: [3, 5, 2, 25, 6, 4, 2, 4, 6, 3, 2]

In [71]:
```
lst.sort()
lst
```
Out[71]: [2, 2, 2, 3, 3, 4, 4, 5, 6, 6, 25]

In [73]:
```
lst.sort(reverse=True)
lst
```
Out[73]: [25, 6, 6, 5, 4, 4, 3, 3, 2, 2, 2]

In [74]:
```
l = lst.copy()
l
```
Out[74]: [25, 6, 6, 5, 4, 4, 3, 3, 2, 2, 2]

In [75]:
```
lst
```

Out[75]: [25, 6, 6, 5, 4, 4, 3, 3, 2, 2, 2]

In [76]:
```
l
```

Out[76]: [25, 6, 6, 5, 4, 4, 3, 3, 2, 2, 2]

In [77]:
```
l.clear()
l
```

Out[77]: []

In [78]:
```
lst
```

Out[78]: [25, 6, 6, 5, 4, 4, 3, 3, 2, 2, 2]

In [80]:
```
lst.count(4)
```

Out[80]: 2

In [81]:
```
lst.index(25)
```

Out[81]: 0

In [ ]:
```

```

Indexing and Slicing

In [82]:
```
lst
```

Out[82]: [25, 6, 6, 5, 4, 4, 3, 3, 2, 2, 2]

In [86]:
```
len(lst)
```

Out[86]:
```
11
```

In [85]:
```
for i in range(len(lst)):
    print(i)
```

```
0
1
2
3
4
5
6
7
8
9
10
```

Indexing

In [87]:
```
lst
```

Out[87]:
```
[25, 6, 6, 5, 4, 4, 3, 3, 2, 2, 2]
```

In [88]:
```
lst[3]
```

Out[88]:
```
5
```

In [89]:
```
lst[8]
```

Out[89]:
```
2
```

In [90]:
```
lst[-3]
```
Out[90]: 2

In [ ]:
```

```

Slicing

In [92]:
```
lst
```
Out[92]: [25, 6, 6, 5, 4, 4, 3, 3, 2, 2, 2]

In [95]:
```
lst[1:6]
```
Out[95]: [6, 6, 5, 4, 4]

In [98]:
```
lst[8:11]
```
Out[98]: [2, 2, 2]

In [99]:
```
lst[8:]
```
Out[99]: [2, 2, 2]

In [106]:
```
lst[-3:]
```
Out[106]: [2, 2, 2]

In [107]:
```
lst[-7:-3]
```
Out[107]: [4, 4, 3, 3]

In [108]:
```
# Steps

lst
```
Out[108]: [25, 6, 6, 5, 4, 4, 3, 3, 2, 2, 2]

In [109]:
```
lst[1:8]
```
Out[109]: [6, 6, 5, 4, 4, 3, 3]

In [110]:
```
lst[1:8:1]
```
Out[110]: [6, 6, 5, 4, 4, 3, 3]

In [111]:
```
lst[1:8:2]
```
Out[111]: [6, 5, 4, 3]

In [112]:
```
lst[1:8:3]
```
Out[112]: [6, 4, 3]

In [113]:
```
lst
```
Out[113]: [25, 6, 6, 5, 4, 4, 3, 3, 2, 2, 2]

In [114]:
```
lst[::-1]
```
Out[114]: [2, 2, 2, 3, 3, 4, 4, 5, 6, 6, 25]

In [115]:
```
lst[-9:-1]
```
Out[115]: [6, 5, 4, 4, 3, 3, 2, 2]

In [116]:
```
lst[-9:-1:2]
```
Out[116]:
```
[6, 4, 3, 2]
```
In [ ]:
```

```

Tuple

In [117]:
```
tup = (2,3,5,6,4,5)
tup
```
Out[117]:
```
(2, 3, 5, 6, 4, 5)
```
In [118]:
```
type(tup)
```
Out[118]:
```
tuple
```
In [119]:
```
tup.append(3)
```
```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[119], line 1
----> 1 tup.append(3)

AttributeError: 'tuple' object has no attribute 'append'
```

In [120]:
```
tup.pop()
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[120], line 1
----> 1 tup.pop()

AttributeError: 'tuple' object has no attribute 'pop'
```

In [121]:
```
tup.remove()
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[121], line 1
----> 1 tup.remove()

AttributeError: 'tuple' object has no attribute 'remove'
```

In [122]:
```
tup
```

Out[122]: (2, 3, 5, 6, 4, 5)

In [123]:
```
tup.count(5)
```

Out[123]: 2

In [124]:
```
tup.index(6)
```

Out[124]: 3

In [125]:
```
tup
```

Out[125]: (2, 3, 5, 6, 4, 5)

In [126]:
```
tup.sort()
```

```
---------------------------------------------------------------------------
AttributeError                            Traceback (most recent call last)
Cell In[126], line 1
----> 1 tup.sort()

AttributeError: 'tuple' object has no attribute 'sort'
```

In [127]:
```
len(tup)
```

Out[127]: 6

In [ ]:

In [ ]:

Dictionary

In [ ]:
```
# { key: value }
```

In [128]:
```
dictt = {'name':'Sourav'}
dictt
```

Out[128]: {'name': 'Sourav'}

In [129]:
```
type(dictt)
```

Out[129]: dict

In [130]:
```
dictt = {

    'name': 'Sourav',
    'age': 26,
    'course': 'Data Science'

}

dictt
```

Out[130]:
```
{'name': 'Sourav', 'age': 26, 'course': 'Data Science'}
```

In [131]:
```
dictt.keys()
```

Out[131]:
```
dict_keys(['name', 'age', 'course'])
```

In [132]:
```
dictt.values()
```

Out[132]:
```
dict_values(['Sourav', 26, 'Data Science'])
```

In [133]:
```
dictt.items()
```

Out[133]:
```
dict_items([('name', 'Sourav'), ('age', 26), ('course', 'Data Science')])
```

In [135]:
```
dictt
```

Out[135]:
```
{'name': 'Sourav', 'age': 26, 'course': 'Data Science'}
```

In [136]:
```
dictt['city'] = 'NOIDA'
```

In [137]:
```
dictt
```

Out[137]:
```
{'name': 'Sourav', 'age': 26, 'course': 'Data Science', 'city': 'NOIDA'}
```

In [138]:
```
dict1 = {

    'name': ['Sourav', 'Himanshu', 'Mohit', 'Rashmi', 'Raj'],
    'age':  [26, 23, 28, 21, 24],
    'city': ['Noida', 'Lucknow', 'Gurgaon', 'Delhi', 'Mumbai']
}

dict1
```

Out[138]:
```
{'name': ['Sourav', 'Himanshu', 'Mohit', 'Rashmi', 'Raj'],
 'age': [26, 23, 28, 21, 24],
 'city': ['Noida', 'Lucknow', 'Gurgaon', 'Delhi', 'Mumbai']}
```

In [139]:
```
dict1['ratings'] = [5, 9, 8, 7, 6, 10]
```

In [140]:
```
dict1
```

Out[140]:
```
{'name': ['Sourav', 'Himanshu', 'Mohit', 'Rashmi', 'Raj'],
 'age': [26, 23, 28, 21, 24],
 'city': ['Noida', 'Lucknow', 'Gurgaon', 'Delhi', 'Mumbai'],
 'ratings': [5, 9, 8, 7, 6, 10]}
```

In [141]:
```
dict1.keys()
```

Out[141]:
```
dict_keys(['name', 'age', 'city', 'ratings'])
```

In [142]:
```
dict1.values()
```

Out[142]:
```
dict_values([['Sourav', 'Himanshu', 'Mohit', 'Rashmi', 'Raj'], [26, 23, 28, 21, 24], ['Noida', 'Luckno
w', 'Gurgaon', 'Delhi', 'Mumbai'], [5, 9, 8, 7, 6, 10]])
```

In [143]:
```
dict1
```

Out[143]:
```
{'name': ['Sourav', 'Himanshu', 'Mohit', 'Rashmi', 'Raj'],
 'age': [26, 23, 28, 21, 24],
 'city': ['Noida', 'Lucknow', 'Gurgaon', 'Delhi', 'Mumbai'],
 'ratings': [5, 9, 8, 7, 6, 10]}
```

In [144]:
```
del dict1['age']
```

In [145]:
```
dict1
```

Out[145]:
```
{'name': ['Sourav', 'Himanshu', 'Mohit', 'Rashmi', 'Raj'],
 'city': ['Noida', 'Lucknow', 'Gurgaon', 'Delhi', 'Mumbai'],
 'ratings': [5, 9, 8, 7, 6, 10]}
```

In [ ]:
```

```

Set Data Types

In [146]:
```
sett = {2, 3, 6, 4, 2, 4, 6}
sett
```

Out[146]:
```
{2, 3, 4, 6}
```

In [147]:
```
sett = {6,3,4,1,8,5}
sett
```

Out[147]:
```
{1, 3, 4, 5, 6, 8}
```

In [148]:
```
type(sett)
```

Out[148]:
```
set
```

In [149]:
```
sett
```

Out[149]:
```
{1, 3, 4, 5, 6, 8}
```

In [150]:
```
sett.add(155)
```

In [151]:
```
sett
```
Out[151]: {1, 3, 4, 5, 6, 8, 155}

In [152]:
```
sett.add('sourav')
sett
```
Out[152]: {1, 155, 3, 4, 5, 6, 8, 'sourav'}

In [154]:
```
sett.add(False)
sett
```
Out[154]: {1, 155, 3, 4, 5, 6, 8, False, 'sourav'}

In [155]:
```
sett.add(34.67)
sett
```
Out[155]: {1, 155, 3, 34.67, 4, 5, 6, 8, False, 'sourav'}

In [157]:
```
sett.add('True')
sett
```
Out[157]: {1, 155, 3, 34.67, 4, 5, 6, 8, False, 'True', 'sourav'}

In [ ]:
```

```

In [158]:
```
sett.discard('True')
sett
```
Out[158]: {1, 155, 3, 34.67, 4, 5, 6, 8, False, 'sourav'}

In [159]:
```
sett.discard('sourav')
sett
```
Out[159]: {False, 1, 3, 4, 5, 6, 8, 34.67, 155}

In [ ]:

In [160]:
```
sett.pop()
sett
```
Out[160]: {1, 3, 4, 5, 6, 8, 34.67, 155}

In [ ]:

In [161]:
```
sett.pop()
sett
```
Out[161]: {3, 4, 5, 6, 8, 34.67, 155}

In [162]:
```
sett.pop()
sett
```
Out[162]: {4, 5, 6, 8, 34.67, 155}

In [ ]:

In [163]:
```
sett.remove(34.67)
sett
```
Out[163]: {4, 5, 6, 8, 155}

In [ ]:

In [164]:
```
sett
```
Out[164]: {4, 5, 6, 8, 155}

In [165]:
```
sett.intersection({5,9})
```
Out[165]:
```
{5}
```

In [ ]:

In [166]:
```
sett.union({5,9})
```
Out[166]:
```
{4, 5, 6, 8, 9, 155}
```

In [ ]:

In [167]:
```
sett.update({2,4,6,7,4.6,1.2, 'a', 'b'})
```

In [168]:
```
sett
```
Out[168]:
```
{1.2, 155, 2, 4, 4.6, 5, 6, 7, 8, 'a', 'b'}
```

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

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